



Implementation of an AI-based Diagnostic Performance Feedback System for Medicine Residents

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Disclosures

- None

Diagnostic Errors Remain a Pervasive Problem

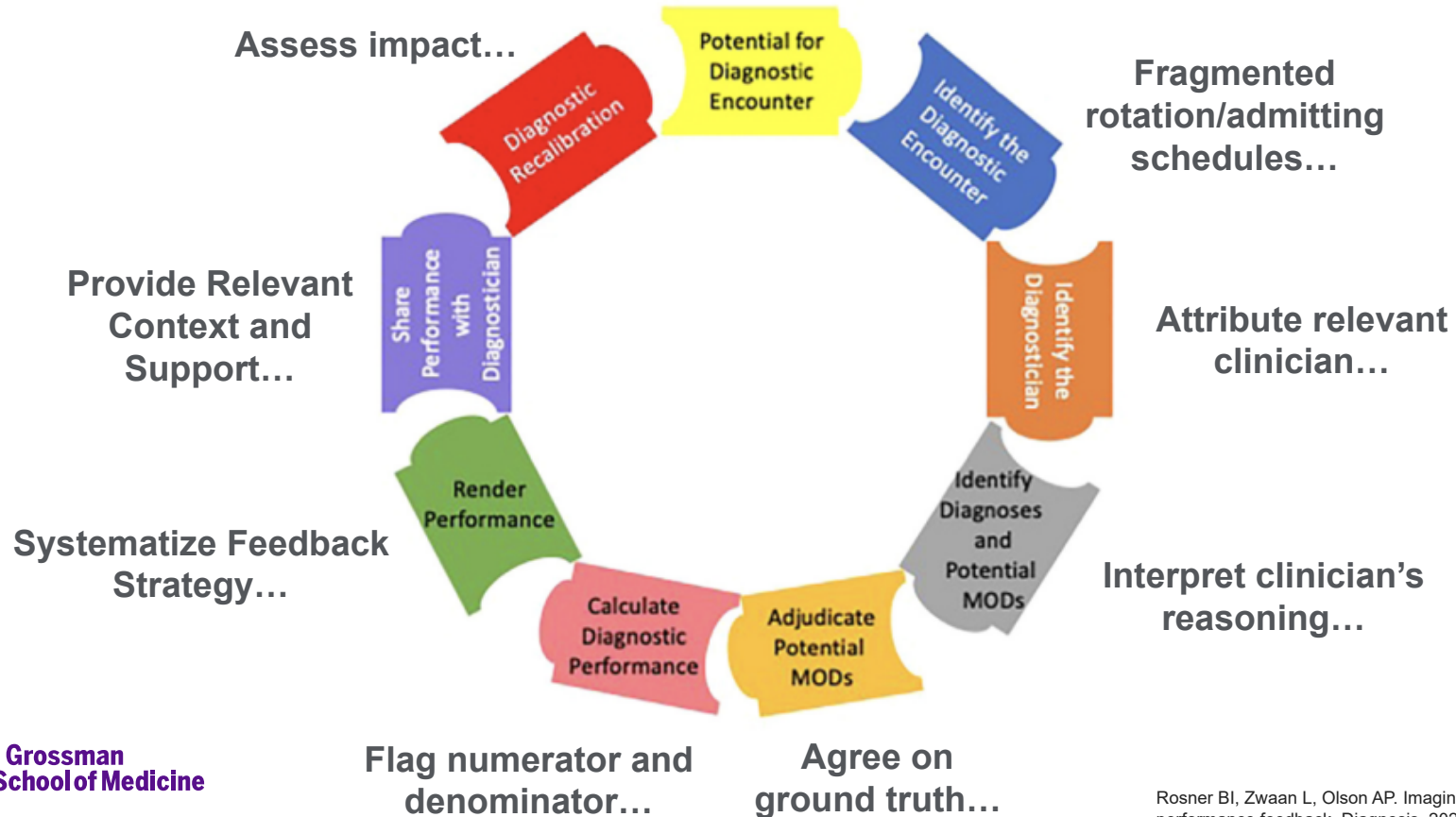
Learning from Errors is Central to Our Mission



Diagnostic Error: “The failure to establish an accurate and timely explanation of a patient’s health problem and communicate that explanation to the patient...”

Call to Action: How can we devise new educational strategies to mitigate error?

Diagnostic Performance Feedback is Critical ... but Challenging



Our Goal

To create an AI-based system that identifies cases of potential diagnostic error in one pilot disease state – venous thromboembolism – and develop a mechanism to provide precise, individualized feedback for internal medicine residents

Basic Idea – Starting with VTE as pilot condition

Hospital Admission Note

Diagnostic Consideration

Actual Diagnosis

Diagnostic Accuracy



✓ Pulmonary Embolism
✗ Pneumonia



✓ Pulmonary Embolism



Accurate



✗ Heart failure
✗ Pneumonia



✓ Pulmonary Embolism

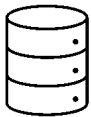


Diagnostic Delay

Discrepancy between diagnostic consideration in the admission note and final diagnosis constitutes a potential diagnostic delay

Retrospective Validation Approach

Pull Retrospective Data



NYULH
Electronic
Health Records

Apply LLM Prompts



Did a resident consider a VTE in their differential diagnosis?



Was a VTE confirmed in radiology imaging reports within 7 days of admission?

Find Discordant Cases

If a resident did NOT consider VTE as potential diagnosis

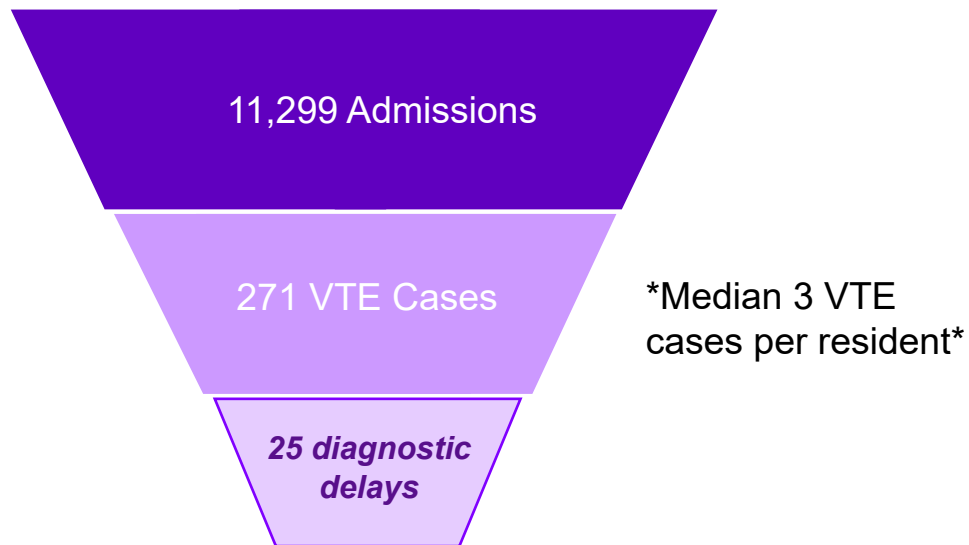
AND

VTE was found in imaging reports

Human SaferDx Reviews

Expert Review to label cases as true Diagnostic Delay, in-hospital complication, or false positive

Case Breakdown During Review Period (2023-2025)



Putting the Data to Work: Resident Advisory Panel



Lauren Heery, MD



Bijal Rajput, MD

- ***Goal: Co-create diagnostic performance feedback platform***
 - Provide feedback on data elements
 - Pilot test and refine early versions of dashboard
 - Guide integration into the residency curriculum
 - Develop expertise in medical education innovations

Diagnostic Performance Dashboard

VTE Diagnoses

Details of Each VTE Diagnosis

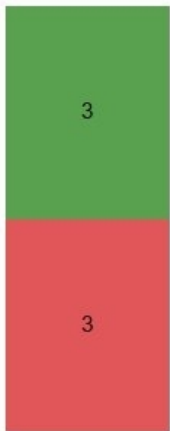
■ Delayed ■ Not Delayed

Patient Characteristics							Contextual Factors					Outcomes					
Resident Name	CSN	Admission Date	Age	Sex	Comorbid ities	Preferred Language	TX Team	Shift	Surge	Admission Number	Days to Dx	Mortality	ICU transfer	RRT	Diagnostic Delay	Differential Note Quality	Explanation Note Quality
Resident A	950077885	1/18/2024	43	Female	moderate	English	MED LB 45C	day	No	1	4	No	No	Yes	Yes	0	0
	976923317	7/23/2024	46	Female	high	Chinese - Cantonese	MED LB 49C	night	No	6	0	No	Yes	No	Yes	2	2
	984337121	9/11/2024	88	Female	moderate	English	MED LB 45B	day	No	1	0	No	No	No	Yes	2	2
	1017334784	4/13/2025	77	Female	low	English	MED LB 49B	day	No	1	0	No	No	No	No	2	2
	1021794766	5/9/2025	87	Female	high	Russian	MED LB 55A	night	No	4	0	No	No	No	No	2	2
	1022198765	5/13/2025	86	Male	high	English	MED LB Oncology	night	No	3	0	No	No	No	No	2	2

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Delayed Not Delayed



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CTPE Positivity Rate
14.3% (You)
vs
14.0% (Peer)



Integrating Performance Feedback into our Curriculum

Introduction to Diagnostic
Performance Dashboard
(n=112, PGY2s/3s)



Independent Guided Exercise
(n=66)

- Perceived Diagnostic Calibration
- Reflective Practice
- Feedback Environment

INTRODUCTION



Providing feedback regarding our ability to generate accurate and timely diagnoses is critical to improving our diagnostic performance. Despite this, it is challenging to review all patients we admit and compare the diagnoses we considered on admission to the actual diagnosis made during a hospitalization.

In this exercise you will have an opportunity to review and reflect on your recent inpatient admissions and work with an AI-based tool that flags potential delayed diagnoses in one condition - venous thromboembolism (VTE). This exercise is an opportunity to reflect on potential diagnostic delays as well as examples of diagnostic excellence from your own practice; this is a formative feedback exercise for you, and is not being used for any other evaluative purpose.

Please answer the first few questions and then log in to your Diagnostic Performance Dashboard to complete the remaining questions. The instructions will guide you through how to log in.

These data are being collected for program evaluation and quality improvement purposes; we will only use these confidential data in aggregate to refine this dashboard and our educational program.

2) Please select your PGY level

- ☐ PGY1
- ☐ PGY2
- ☐ PGY3

Perceived Impact on Calibration and Reflection

Perceived Impact on Calibration and Reflection

Question	Total (n=66)
	Somewhat or a great deal, % (n)
Does reviewing your diagnostic performance in VTE through this exercise give you insight into your diagnostic calibration?	59%, (39)

Perceived Impact on Calibration and Reflection

Question	Total (n=66)	Campus		
	Somewhat or a great deal, % (n)	Brooklyn (n=30)	Manhattan (n=36)	P-value
Does reviewing your diagnostic performance in VTE through this exercise give you insight into your diagnostic calibration?	59%, (39)	73%, (22)	47%, (17)	p=0.03

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Does reviewing your diagnostic performance in VTE through this exercise give you insight into your diagnostic calibration?	59%, (39)	73%, (22)	47%, (17)	p=0.03
After completing this activity, to what extent do you plan to follow up on patient outcomes after leaving your care?	48%, (32)	70%, (21)	30%, (11)	p=0.001

Perceived Impact on Calibration and Reflection

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After completing this activity, to what extent do you plan to follow up on patient outcomes after leaving your care?	48%, (32)	70%, (21)	30%, (11)	p=0.001
Has this exercise encouraged you to reflect on your diagnostic performance in other conditions?	89%, (59)	93%, (28)	86%, (31)	p=0.34

There are campus-specific differences in perceived impact of this AI-based feedback platform

The Different Campuses

NYU Internal Medicine

Brooklyn



We capture the entirety of
Brooklyn residents' encounters

Manhattan



Bellevue

VA

U.S. Department
of Veterans Affairs

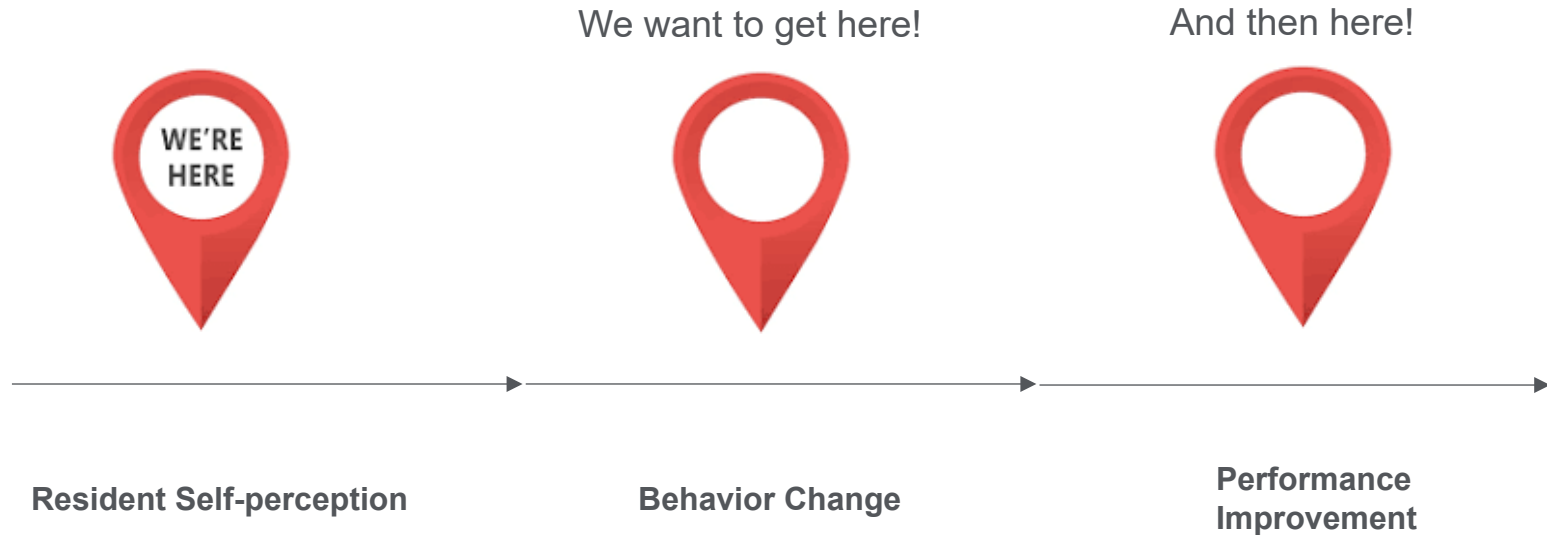
We capture a subset of
Manhattan residents' encounters

Feedback Environment Fostered by AI-based Platform

How is receiving diagnostic feedback through this dashboard different?

Themes	Representative Quote
Data-Driven Insights Foster Comfort with Self-Reflection	"This data is very objective, and revolves around self-reflection. In this regard, criticism can feel less targeted but encourages me to evaluate my own performance and seek improvement."
Independent and Self-Paced Feedback Allows for Deeper Reflection	"It is nice to get feedback in a way that I could spend as much time reviewing cases as I want and contemplating how I could have changed my work-up/management. It is also nice to get feedback on cases when I feel ready to receive/review feedback."
Lack of Context and Nuance Compared to Supervisor Feedback	"The dashboard provides objective, data-driven feedback that feels less personal than supervisor evaluations. That said, it lacks the context and nuance a supervisor can provide—for example, understanding patient complexity, workflow constraints, or team dynamics that shaped certain decisions."

Spectrum of Impact of this Intervention

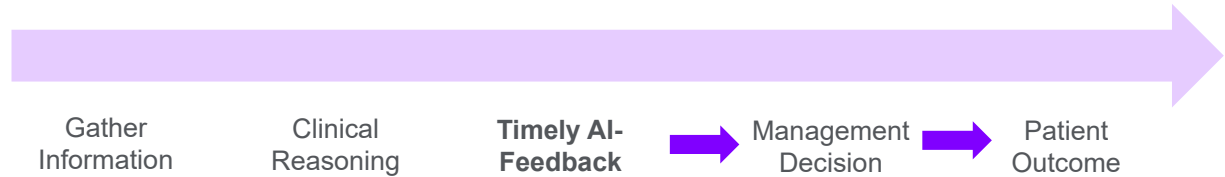


What's next?

Feedback on Action



Feedback in Action



The Team



Verity Schaye, MD, MHPE



Jesse Burk-Rafel, MD



Bijal Rajput, MD



Lexi Signoriello, PhD

Funding



Joseph and Diane Steinberg

Questions?